USN | | | | | | | | | | | [23BSMA13]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024
CALCULUS, LAPLACE TRANSFORMS AND MATRICES

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Missing data may be suitably assumed.
- **ii.** Answer all questions

		PART - A			**05 × 02=10**
1.	**a.**	If $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ then find $\frac{dy}{dx}$ using partial derivatives.	L-2	CO1	**02**
	b.	If $\vec{v} = yz\,\hat{\imath} + 3zx\,\hat{\jmath} + z\,\hat{k}$ then $curl\vec{v}$ ___	L-2	CO1	**02**
	c.	$L[t\cosh at] =$ ___	L-2	CO3	**02**
	d.	$L^{-1}\left[\frac{1+e^{-3s}}{s^2}\right] =$ ___	L-2	CO4	**02**
	e.	Find the rank of the matrix $A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$	L-2	CO1	**02**
		PART - B			**06 × 05=30**
2.	**a.**	If $z = f(x+ct) + \varphi(x-ct)$ then prove that $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$	L-3	CO2	**05**
		OR			
	b.	If $u = f(x-y,\ y-z,\ z-x)$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.	L-3	CO2	**05**
3.	**a.**	Find $div\,\vec{F}$ and $curl\,\vec{F}$ where $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$.	L-3	CO6	**05**
		OR			
	b.	If $x = t^2 + 1, y = 4t - 3, z = 2t^2 - 6t$ represents the parametric equation of a curve, then find **i.** The unit tangent vector at any point on the curve **ii.** The angle between the tangents at t=1 and t=2.	L-3	CO1	**05**
4.	**a.**	Evaluate $L\left[\frac{1-e^{-at}}{t}\right]$.	L-3	CO3	**05**
		OR			
	b.	Express $f(t) = \begin{cases} t, & 0 < t < 4 \\ 5, & t > 4 \end{cases}$ in terms of unit step function and hence find its Laplace transform.	L-3	CO3	**05**
5.	**a.**	Evaluate $L^{-1}\left[\log\left(1 - \frac{a^2}{s^2}\right)\right]$.	L-3	CO4	**05**

		OR			
	b.	Find the inverse Laplace Transform of $\frac{s+5}{s^2-6s+13}$.	L-3	CO4	**05**
6.	**a.**	Identify the rank of the following matrix by reducing into the row echelon form $A = \begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$	L-3	CO1	**05**
		OR			
	b.	Investigate the Eigen values and the corresponding Eigen vectors of the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$.	L-3	CO5	**05**
7.	**a.**	Write MATLAB code to find the first and second-order partial derivatives of the function $f(x,y) = x^2 + 5xy - 2y^2$.	L-3	CO2	**05**
		OR			
	b.	Write MATLAB code to determine the Inverse Laplace transform for the following functions **i.** $\frac{1}{(s-a)^2+b^2}$ **ii.** $\frac{3s-8}{4s^2+25}$ **iii.** $\frac{3s-1}{s^2+3s+2}$.	L-3	CO1	**05**
		PART - C		**06 × 10=60**	
8.	**a.**	If $x = e^u cosv$ and $y = e^u sinv$ then show that $JJ' = 1$	L-3	CO2	**10**
		OR			
	b.	Prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{(x+y+z)}$, where $u = log(x^3 + y^3 + z^3 - 3xyz)$ and hence show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$.	L-3	CO2	**10**
9.	**a.**	Show that $\vec{F} = \frac{x\,i + y\,j}{x^2 + y^2}$ is both solenoidal and irrotational.	L-3	CO6	**10**
		OR			
	b.	Show that $\vec{F} = (2xy^2 + yz)i + (2x^2y + xz + 2yz^2)j + (2y^2z + xy)k$ is conservative force field. Find its scalar potential.	L-3	CO6	**10**
10.	**a.**	Show that $\int_0^\infty t^3 e^{-t} sint = 0$.	L-3	CO3	**10**
		OR			
	b.	Given $f(t) = \begin{cases} E, & 0 < t < a/2 \\ -E, & a/2 < t < a \end{cases}$ where $f(t+a) = f(t)$, show that	L-3	CO3	**10**

		$L[f(t)] = \frac{E}{s} . tanh(as/4)$.			
11.	**a.**	Solve $y''' + 2y'' - y' - 2y = 0$ given $y(0) = y'(0) = 0$ and $y''(0) = 6$ by using Laplce transform.	L-3	CO4	**10**
		OR			
	b.	**i.** State convolution theorem. **ii.** Using convolution theorem find the inverse Laplace transform of $\frac{s}{(s^2+a^2)^2}$.	L-3	CO4	**10**
12.	**a.**	Solve the system of equations by Gauss-Seidel method $10x + y + z = 12$, $x + 10y + z = 12$ and $x + y + 10z = 12$.	L-3	CO1	**10**
		OR			
	b.	Investigate the values of λ and μ so that the system of equations $x + y + z = 6$, $x + 2y + 3z = 10$ and $x + 2y + \lambda z = \mu$ will have **i.** no solution **ii.** a unique solution **iii.** infinite number of solutions	L-3	CO1	**10**
13.	**a.**	Write MATLAB code to draw two-dimensional vector field for the vector $x\hat{i} + y\hat{j}$.	L-3	CO4	**10**
		OR			
	b.	Write MATLAB code to generate matrix. Perform the following operation for matrix $A = \begin{bmatrix} 1 & -2 & -3 \\ -2 & 2 & 5 \\ -3 & 5 & 3 \end{bmatrix}$ using MATLAB **i.** Diagonal of a matrix A **ii.** Size of matrix A **iii.** Rank of A **iv.** Inverse of a matrix A **v.** Characteristic polynomial of the square matrix A	L-3	CO1	**10**